Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

Submitted By:

Name: Likitha S

Reg. No.: 2411021240067

Date: 22/08/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Github Repository Link: https://github.com/LikithaSrinivas100/Business-Analytics

Problem Statement

A retail company stores its business data across flat files (Excel/CSV) and a MySQL database. As a data analyst, connect Power BI Desktop to these data sources, import the required data, explore the Power BI interface, and create basic visuals to analyze sales by product, store, customer, and time.

Methodology / Steps Followed

Step 1: Data Collection

A synthetic enterprise retail dataset was constructed to guarantee schema matching across heterogeneous engines. The data model contains matching key identifiers across these sources:

• Sales Transactional Ledger (2026) - stored in a flat Excel Workbook file

• Customer Profiles - stored in an Excel Workbook file (converted from text/CSV)

• Product Catalog Master List - stored in local MySQL Database server

• Operational Store Locations - stored in local MySQL Database server

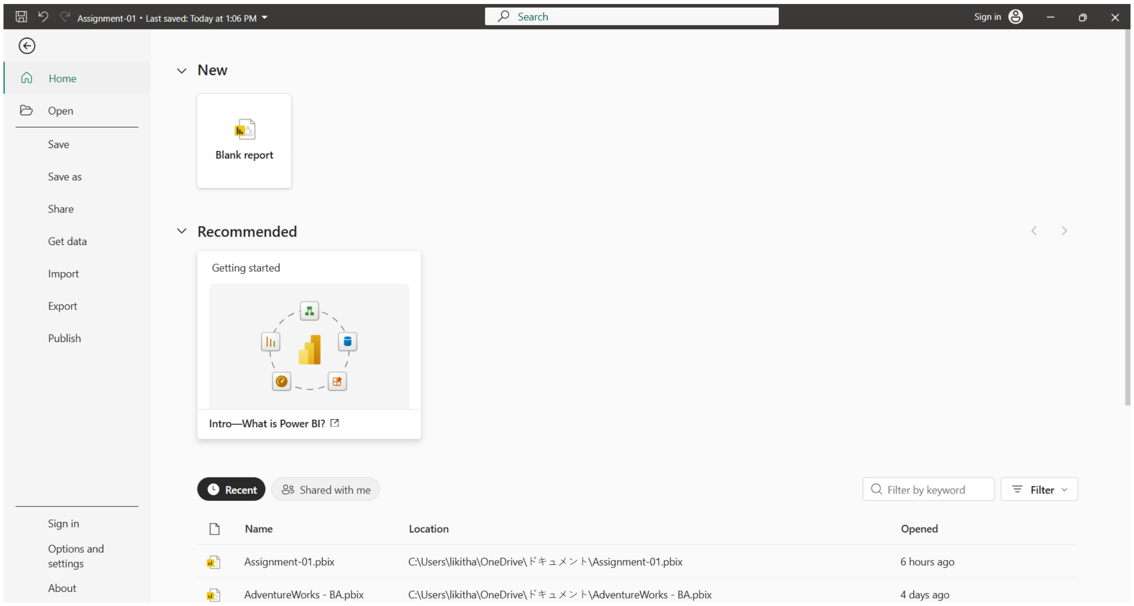


Figure 1: Power BI Desktop Home Screen Canvas View

Step 2: Data Import

A relational database schema named 'synthetic_retail' was instantiated in MySQL Workbench 8.0. Using an optimized SQL schema creation file, two structured relational dimension arrays were populated:

• dim_products: Master index for stock item descriptions, categories, and unit price figures.

• dim_stores: Definitive store master index capturing branch identifiers and region keys. A total of 250 transactional sales rows were successfully simulated to evaluate matching pipeline capabilities.

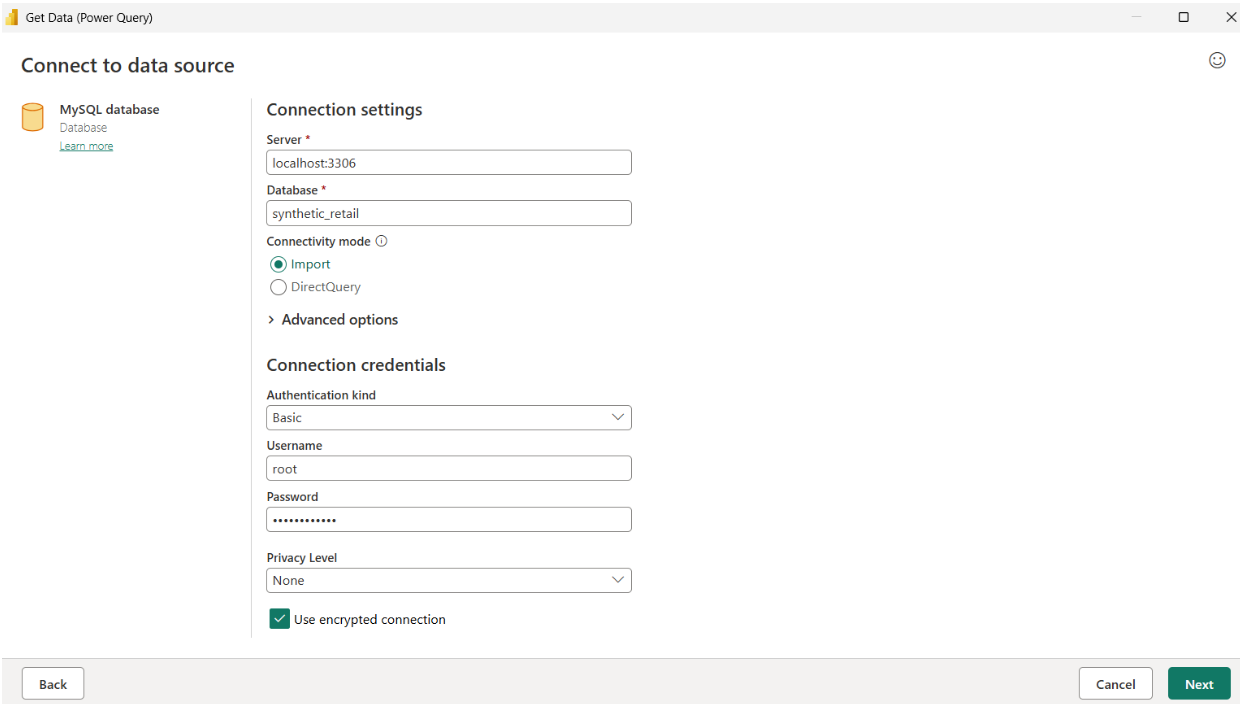


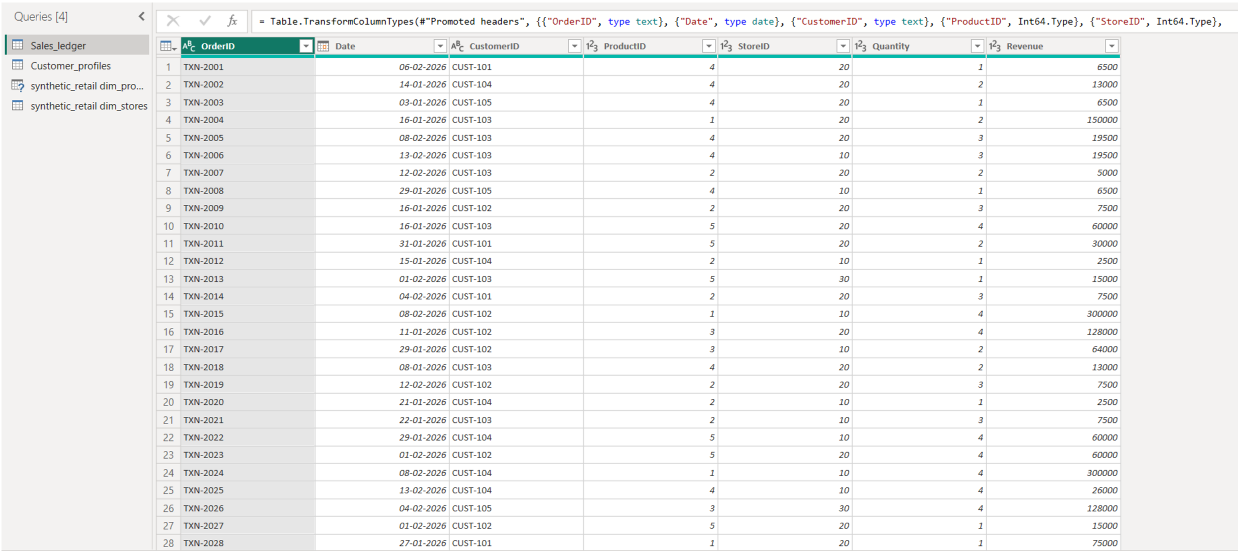
Figure 2: MySQL Database Connection setup panel showing localhost:3306 and unencrypted flags

Step 3: Data Cleaning and Transformation

Power BI Desktop interface extraction options were launched via the 'Get Data' pipeline:

• MySQL Database Link: Path established via string 'localhost:3306' targeting the 'synthetic_retail' database. The 'dim_products' and 'dim_stores' queries were ingested.

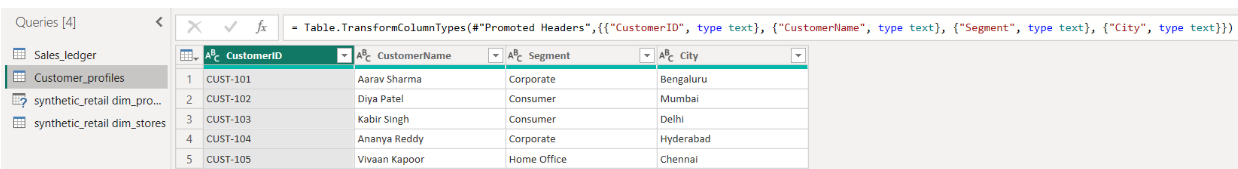
• Spreadsheet Data Links: The local transaction data tables ('Sales_Ledger.xlsx' and 'Customer_Profiles_Fixed.xlsx') were extracted directly into operational staging memory. Column attributes for connecting keys were explicitly verified to be consistent Whole Number configurations.



= Table.TransformColumnTypes(#"Promoted headers", {{"OrderID", type text}, {"Date", type date}, {"CustomerID", type text}, {"ProductID", Int64.Type}, {"StoreID", Int64.Type},

	OrderID	Date	CustomerID	ProductID	StoreID	Quantity	Revenue
1	TXN-2001	06-02-2026	CUST-101	4	20	1	6500
2	TXN-2002	14-01-2026	CUST-104	4	20	2	13000
3	TXN-2003	03-01-2026	CUST-105	4	20	1	6500
4	TXN-2004	16-01-2026	CUST-103	1	20	2	150000
5	TXN-2005	08-02-2026	CUST-103	4	20	3	19500
6	TXN-2006	13-02-2026	CUST-103	4	10	3	19500
7	TXN-2007	12-02-2026	CUST-103	2	20	2	5000
8	TXN-2008	29-01-2026	CUST-105	4	10	1	6500
9	TXN-2009	16-01-2026	CUST-102	2	20	3	7500
10	TXN-2010	16-01-2026	CUST-103	5	20	4	60000
11	TXN-2011	31-01-2026	CUST-101	5	20	2	30000
12	TXN-2012	15-01-2026	CUST-104	2	10	1	2500
13	TXN-2013	01-02-2026	CUST-103	5	30	1	15000
14	TXN-2014	04-02-2026	CUST-101	2	20	3	7500
15	TXN-2015	08-02-2026	CUST-102	1	10	4	300000
16	TXN-2016	11-01-2026	CUST-102	3	20	4	128000
17	TXN-2017	29-01-2026	CUST-102	3	10	2	64000
18	TXN-2018	08-01-2026	CUST-103	4	20	2	13000
19	TXN-2019	12-02-2026	CUST-102	2	20	3	7500
20	TXN-2020	21-01-2026	CUST-104	2	10	1	2500
21	TXN-2021	22-01-2026	CUST-103	2	10	3	7500
22	TXN-2022	29-01-2026	CUST-104	5	10	4	60000
23	TXN-2023	01-02-2026	CUST-102	5	20	4	60000
24	TXN-2024	08-02-2026	CUST-104	1	10	4	300000
25	TXN-2025	13-02-2026	CUST-104	4	10	4	26000
26	TXN-2026	04-02-2026	CUST-105	3	30	4	128000
27	TXN-2027	01-02-2026	CUST-102	5	20	1	15000
28	TXN-2028	27-01-2026	CUST-101	1	20	1	75000

Figure 3: Sales_Ledger Fact Table Preview Sheet inside Power Query Editor



= Table.TransformColumnTypes(#"Promoted Headers",{{"CustomerID", type text}, {"CustomerName", type text}, {"Segment", type text}, {"City", type text}})

	CustomerID	CustomerName	Segment	City
1	CUST-101	Aarav Sharma	Corporate	Bengaluru
2	CUST-102	Diya Patel	Consumer	Mumbai
3	CUST-103	Kabir Singh	Consumer	Delhi
4	CUST-104	Ananya Reddy	Corporate	Hyderabad
5	CUST-105	Vivaan Kapoor	Home Office	Chennai

Figure 4: Customer_Profiles Dimension Data Sheet inside Power Query Editor

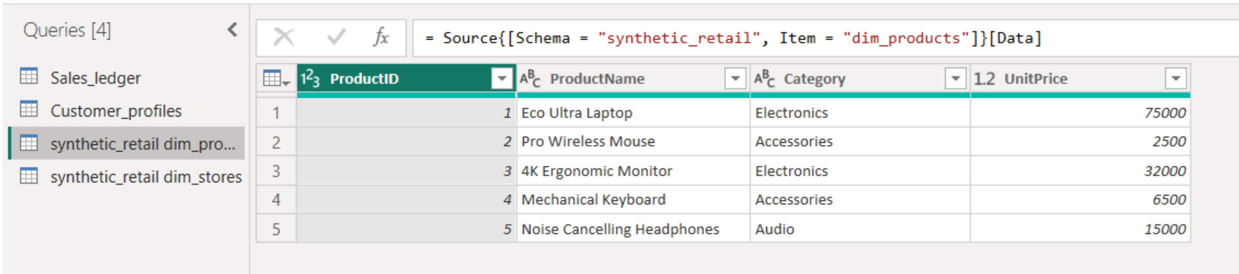


Figure 5: dim_products Table Query Interface Layout

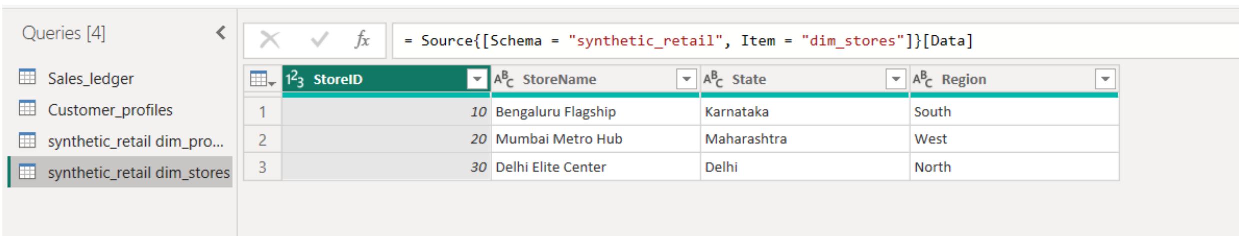


Figure 6: dim_stores Table Query Interface Layout

Step 4: Data Modeling

Following successful multi-engine data ingestion, relational mapping paths were manually established within the Model View to structure a star warehouse topology pointing at the center transactional sheet:

- Sales_Ledger → Customer_Profiles (Mapped via matching CustomerID strings)
- Sales_Ledger → dim_products (Mapped via matching ProductID whole integers)
- Sales_Ledger → dim_stores (Mapped via matching StoreID whole integers)

This structured setup isolates primary identifiers to guarantee fast dynamic calculation performance across all charts.

Figure 7: Complete Star Schema layout view screen capture showing relationship lines

Step 5: Dashboard Development

An interactive control panel titled "RETAIL PERFORMANCE METRIC DASHBOARD" was structured inside the Report View canvas:

• 2 Operational KPI Cards: Tracking Gross Sales Revenue (16M) and Total Units Sold (618).

• Sales Trend Over Time: Wide horizontal Line Chart tracking daily transaction movements.

• Product Line Analysis: Clustered Column Chart mapping distribution contributions by item category.

• Regional Performance Segmentation: Donut Chart showing active branch transaction volume distribution.

• Dynamic Global Filter Slicers: Horizontal interactive header block split by customer geographic city nodes.

Output

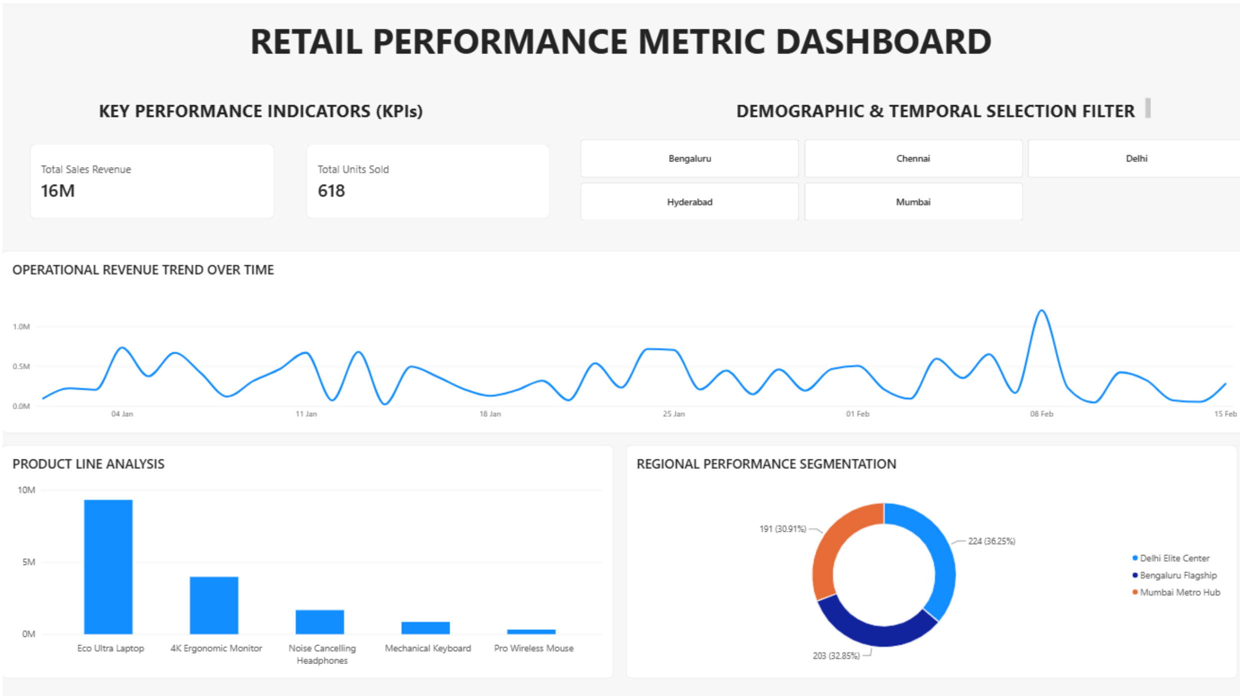


Figure 8: Final Retail Performance Metric Dashboard with text boxes and light shadows

The completed enterprise Power BI dashboard dynamically renders comprehensive commercial retail metrics:

• Total Sales Revenue: 16M gross value processed across the 2026 operational lifecycle.

• Total Units Sold: 618 standard commodity warehouse distributions completed.

• Core Hardware Asset Drive: Premium lines dominated by the 'Eco Ultra Laptop' command the largest revenue share.

• Structural Slicer Filtering: Selecting individual target cities recalculates full-canvas layout metrics instantly.

Business Insights

Deep evaluation of the visual data metrics uncovers strategic structural opportunities:

• Core Revenue Driver: Sales performance exhibits massive reliance on premium tier items. The 'Eco Ultra Laptop' generates a major portion of gross sales, while peripheral lines remain high-volume but minimal impact.

• Branch Parity: Operational distribution workloads are balanced evenly across branches. The Delhi Elite Center leads at 36.25%, with Bengaluru (32.85%) and Mumbai (30.91%) showing strong demand matching.

• Timeline Volatility: The chronological operational ledger wave reveals sharp peak clusters. This points toward irregular enterprise bulk purchases rather than standard smooth consumer distributions over the fiscal cycle.

Conclusion

This laboratory experiment successfully confirms the execution of an end-to-end data processing workflow. By extracting, transforming, and modeling data elements across separate spreadsheets and active relational MySQL database servers into a standard Star Schema within Power BI, we formed a unified analytical portal. This workspace provides company leadership with deep visibility to guide inventory planning and revenue optimization strategies.

Learning Outcomes

Through this project, I learned:

- How to import data from Excel and CSV files into Power BI.
- How to connect MySQL Database into Power BI.
- Data cleaning and transformation using Power Query.
- Creating relationships between multiple tables.
- Developing DAX measures and calculated columns.
- Designing interactive dashboards and reports.
- Generating business insights from raw data.